Python Project
Presentation

Sales Data Analysis

Analyzing Sales Performance with Python, Pandas & Matplotlib

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Objectives

Calculate Total Sales

Sum all sales values to understand overall revenue.

Determine Average Sales

Compute mean sales to benchmark performance.

Identify Highest Sales

Find the maximum sale and top performer.

Analyze Region Performance

Compare sales across North, South, East, West.

Tools & Dataset

Python
Core programming language for data analysis.

Pandas
Library for DataFrame creation and manipulation.

Matplotlib
Visualization library for creating bar charts.

Dataset Structure
Employee names, sales values, and region details.

Methodology Steps

Created dataset using dictionary and converted into DataFrame using Pandas. Calculated total sales, average sales, and highest sales. Identified top performer and performed region-wise analysis using groupby(). Visualized data using bar chart.

Python Code

```
import pandas as pd
data={"Names":["abinaya","sandhiya","ashwini","varsha","farin"],
    "Sales":[100,200,150,300,250],
    "Region":["North","South","East","West","North"]}
df=pd.DataFrame(data)
print(df)
```

Program Output

```
ect.py
    Names  Sales Region
0  abinaya    100  North
1 sandhiya    200  South
2  ashwini    150   East
3   varsha    300   West
4    farin    250  North
```

Key Results & Findings

Total Sales = 1000
Average Sales = 200
Highest Sales = 300
Top Performer: Varsha - 300

Python code

```
File Edit Format Run Options Window Help
import pandas as pd
data={"Names":["abinaya","sandhiya","ashwini","varsha","farin"],
    "Sales":[100,200,150,300,250],
    "Region":["North","South","East","West","North"]}
df=pd.DataFrame(data)
print(df)
print("Total Sales:",df["Sales"].sum())
print("Average Sales:",df["Sales"].mean())
print("Highest Sales:",df["Sales"].max())
print(df[df["Sales"]==df["Sales"].max()])
print("Region wise sales:")
print(df.groupby("Region")["Sales"].sum())
```

Program Output

```
= RESTART: C:\Users\Admin\AppData\Lo
   Names  Sales Region
0  abinaya    100  North
1  sandhiya   200  South
2  ashwini    150  East
3  varsha     300  West
4  farin      250  North
Total Sales: 1000
Average Sales: 200.0
Highest Sales: 300
   Names  Sales Region
3  varsha    300  West
Region wise sales:
Region
East    150
North   350
South   200
West    300
Name: Sales, dtype: int64
```

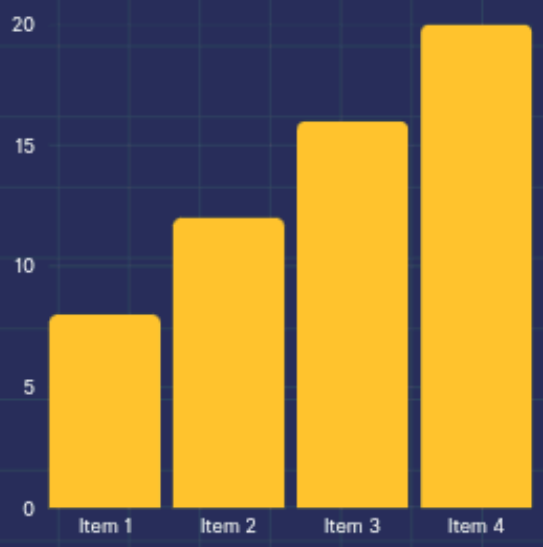
Region-wise Sales

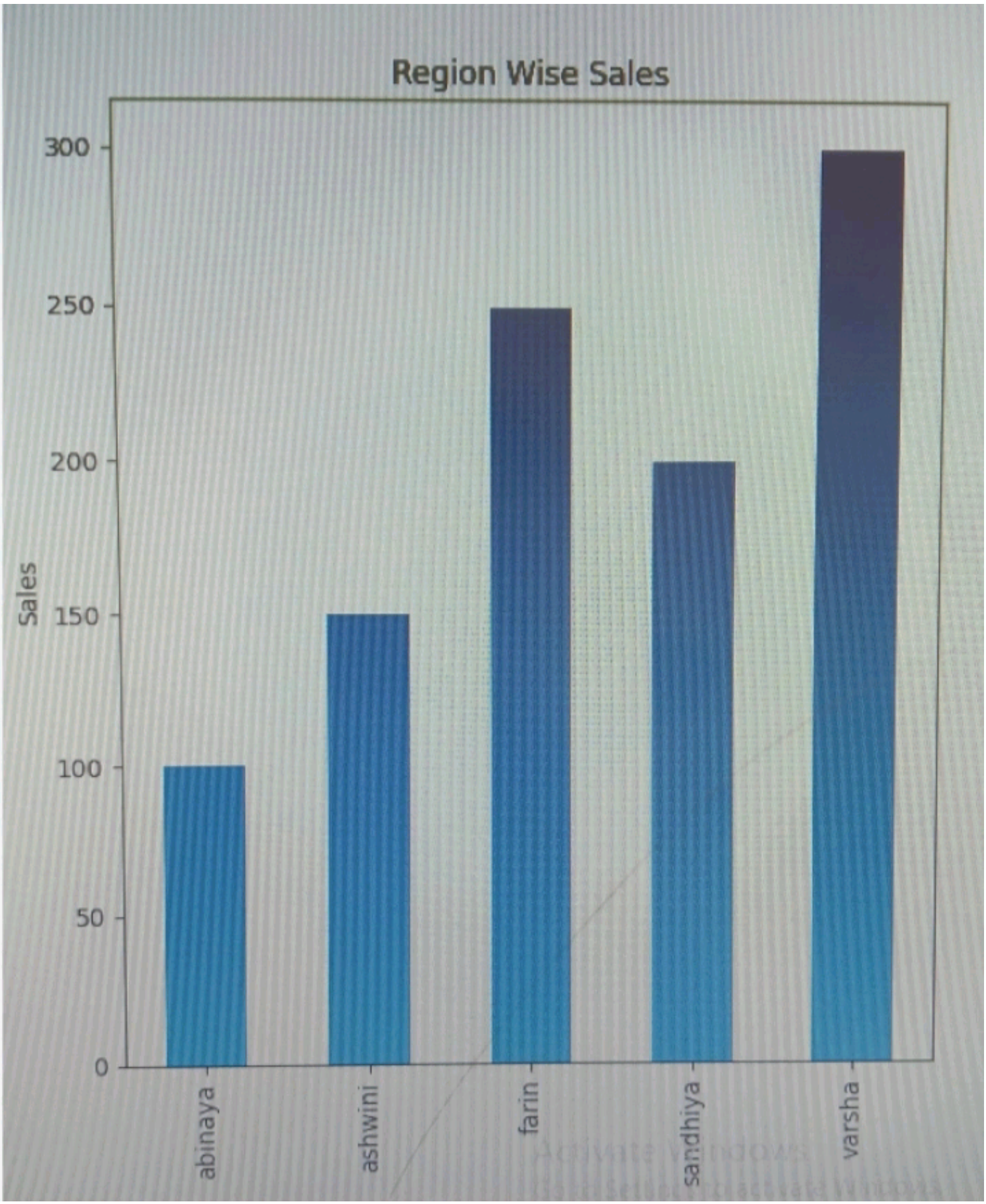
East Region	North Region	South Region	West Region
Total Sales: ₹150	Total Sales: ₹350	Total Sales: ₹200	Total Sales: ₹300

REGION-WISE SALES VISUALIZATION

Bar Chart Analysis

Bar chart used to represent region-wise sales performance. This visualization helps to easily compare performance across different regions at a glance.




Region Wise Sales
Sales
300
250
200
150
100
50
0
abinaya
ashwini
farin
sandhiya
varsha

KEY INSIGHTS
Important Findings from Our Sales Analysis
North region has the highest sales performance
Varsha is the top performer among all employees
East region has the lowest sales numbers
Sales performance varies significantly across regions
Data-driven insights enable strategic decisions

CONCLUSION &
KEY TAKEAWAYS
Understanding Sales Trends
with Python Analytics
North region achieved highest sales per
Varsha identified as top sales performer
East region shows lowest sales metrics
Regional performance varies significantly
Python tools enable data-driven decisions

Sales Data Analysis
Project

Thank You!

Questions? are welcome

Sales Data Analysis using Python